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14.2

18. Three Entrances A study was conducted to aid in the remodeling of an office building that contains three entrances. The choice of entrance was recorded for a sample of 200 persons who entered the building. Do the data in the table indicate that there is a difference in preference for the three entrances? Find a 95% confidence interval for the proportion of persons favoring entrance 1.

Entrance	1	2	3
Number Entering	83	61	56

$n = 200$

$H_0: p_1 = p_2 = p_3 = \frac{1}{3}$ all entrances have same probability

H_a: at least one entrance proportion is different

$O_1 = 83$, $O_2 = 61$, $O_3 = 56$, $n = 200$

$E_i = np_i$

$E_1 = 200\left(\frac{1}{3}\right) = 66.67$

$E_2 = 200\left(\frac{1}{3}\right) = 66.67$

$E_3 = 200\left(\frac{1}{3}\right) = 66.67$

Tstat: $X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

$$X^2 = \frac{(83-66.67)^2}{66.67} + \frac{(61-66.67)^2}{66.67} + \frac{(56-66.67)^2}{66.67}$$

$= 3.99 + 0.48 + 1.71 = 6.18$

3 categories

$df = k - 1 = 3 - 1 = 2$

$\alpha = 0.05 \rightarrow X^2_{0.05, 2} = 5.991$

RR: $X^2 > 5.991$

$6.18 > 5.991$ (Reject H_0)

→ There is sufficient evidence at 0.05 significance level to conclude that the three entrances are not equally preferred

95% CI for entrance 1 proportion

$\hat{p} = \frac{83}{200} = 0.415$

$CI = \hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

$0.415 \pm 1.96 \sqrt{\frac{0.415(0.585)}{200}}$

0.415 ± 0.0683

$(0.347, 0.483)$

22. Heart Attacks on Mondays Researchers from Germany have concluded that the risk of a heart attack for a working person may be as much as 50% greater on Monday than on any other day.[1] In an attempt to verify their claim, 200 working people who had recently had heart attacks were surveyed and the day on which their heart attacks occurred was recorded:

Day	Observed Count
Sunday	24
Monday	36
Tuesday	27
Wednesday	26
Thursday	32
Friday	26
Saturday	29

$n = 200$

Do the data present sufficient evidence to indicate that there is a difference in the incidence of heart attacks depending on the day of the week? Test using $\alpha = .05$.

$H_0: P_{sun} = P_{mon} = P_{tue} = P_{wed} = P_{thu} = P_{fri} = P_{sat} = \frac{1}{7}$

H_a: at least one day has a different proportion

$E_i = n\left(\frac{1}{7}\right) = \frac{200}{7} = 28.571$

tstat: $X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

$$X^2 = \frac{(24-28.571)^2}{28.571} + \frac{(36-28.571)^2}{28.571} + \frac{(27-28.571)^2}{28.571} + \frac{(26-28.571)^2}{28.571} + \frac{(32-28.571)^2}{28.571} + \frac{(26-28.571)^2}{28.571} + \frac{(29-28.571)^2}{28.571}$$

$= 0.731 + 1.932 + 0.086 + 0.231 + 0.412 + 0.231 + 0.006$

$= 3.629$

$df = k - 1 = 7 - 1 = 6$

$\alpha = 0.05 \rightarrow X^2_{0.05, 6} = 12.592$

RR: $X^2 > 12.592$

$3.629 < 12.592$ (Fail to reject H_0)

→ There is no sufficient evidence at 0.05 significance level to conclude that the incidence of heart attacks differs by day of the week.

$H_0: p_1 = p_1 = p_2$

H_a: at least one differs

$E(np_i)$ — probability $\left(\frac{1}{\text{jml prob}}\right)$

total 'n' sample

tstat: $X^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

$df: k - 1$

RR: $X^2 > X^2_{\alpha, df}$

Confidence Interval

$\hat{p} \pm Z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$

14.3 — TWO CATEGORICAL VARIABLES

18. Fatal Accidents Accident data were analyzed to determine the numbers of fatal accidents for automobiles of three sizes. The data for 346 accidents are as follows:

	Small	Medium	Large	total
Fatal	67	26	16	109
Not Fatal	128	63	46	237
Total	195	89	62	346

Do the data indicate that the frequency of fatal accidents is dependent on the size of automobiles? Test using a 5% significance level.

H_0: Fatality and automobile size are independent

H_a: Fatality and automobile size are dependent

$E_{ij} = \frac{(\text{row total})(\text{column total})}{n}$

(FS) $E_{11} = \frac{109 \times 195}{346} = 61.43$ (NS) $E_{21} = \frac{237 \times 195}{346} = 133.57$

(FM) $E_{12} = \frac{109 \times 89}{346} = 28.03$ (NM) $E_{22} = \frac{237 \times 89}{346} = 60.96$

(FL) $E_{13} = \frac{109 \times 62}{346} = 19.53$ (NL) $E_{23} = \frac{237 \times 62}{346} = 42.47$

harus kaliin total & column satu"

tstat: $X^2 = \sum \frac{(O-E)^2}{E}$ pake E di msg" kolom & row

$$X^2 = \frac{(67-61.43)^2}{61.43} + \frac{(26-28.03)^2}{28.03} + \frac{(16-19.53)^2}{19.53} + \frac{(128-133.57)^2}{133.57} + \frac{(63-60.96)^2}{60.96} + \frac{(46-42.47)^2}{42.47}$$

$= 0.505 + 0.147 + 0.638 + 0.232 + 0.068 + 0.293$

$= 1.883$

$df = (\overset{\text{row}}{r} - 1)(\overset{\text{column}}{c} - 1)$

$= (2-1)(3-1) = 2$

$\alpha = 0.05 \rightarrow X^2_{0.05, 2} = 5.991$

RR: $X^2 > 5.991$

$1.883 < 5.991$ (Fail to reject H_0)

→ There is no sufficient evidence at 0.05 significance level to conclude that the frequency of fatal accidents depends on automobile size

DATA SET DS1403

23. Telecommuting As an alternative to flextime, many companies allow employees to do some of their work at home. Individuals in a random sample of 300 workers were classified according to salary and number of workdays per week spent at home.

	Workdays at Home per Week			
Salary	Less Than One	At Least One, but Not All	All at Home	total
Under \$50,000	38	16	14	68
\$50,000–\$74,999	54	26	12	92
\$75,000–\$99,999	35	22	9	66
Above \$100,000	33	29	12	74
total	160	93	47	300 n

a. Do the data present sufficient evidence to indicate that salary is dependent on the number of workdays ? spent at home? Test using $\alpha = .05$.

b. Use Table 5 in Appendix I to approximate the p-value for this test of hypothesis. Does the p-value confirm your conclusions from part a?

a) H_0: salary & number of workdays at home are independent

H_a: salary & number of workdays at home are dependent

Expected table

salary	less than 1	at least 1	all at home
<50k	36.27 $\left[\frac{68 \times 160}{300}\right]$	21.08 $\left[\frac{68 \times 93}{300}\right]$	10.65 $\left[\frac{68 \times 47}{300}\right]$
50k-74.999	49.07 $\left[\frac{92 \times 160}{300}\right]$	28.52 dll.	14.41
75k-99.999	35.20	20.46	10.34
>100k	39.47	22.94	11.59

Values are computed by $E_{ij} = \frac{(\text{row total})(\text{column total})}{n}$

tstat: $X^2 = \sum \frac{(O-E)^2}{E}$

$$= \frac{(38-36.27)^2}{36.27} + \frac{(16-21.08)^2}{21.08} + \frac{(14-10.65)^2}{10.65} + \frac{(54-49.07)^2}{49.07} + \frac{(26-28.52)^2}{28.52} + \frac{(12-14.41)^2}{14.41} + \frac{(35-35.20)^2}{35.20} + \frac{(22-20.46)^2}{20.46} + \frac{(9-10.34)^2}{10.34} + \frac{(33-39.47)^2}{39.47} + \frac{(29-22.94)^2}{22.94}$$

$$+ \frac{(12-11.59)^2}{11.59}$$

$= 6.448$

$df = (r-1)(c-1) = (4-1)(3-1) = 6$

$\alpha = 0.05 \rightarrow X^2_{0.05, 6} = 12.592$

RR: $X^2 > 12.592$

$6.448 < 12.592$ (Fail to reject H_0)

→ There is no sufficient evidence at 0.05 significance level to conclude that salary is dependent on number of workdays spent at home

b) $df = 6$

$X^2 = 5.49$

$p \approx 0.48$ since $p > 0.05$

→ Yes, the p-value confirms the conclusion that we do not reject H_0.